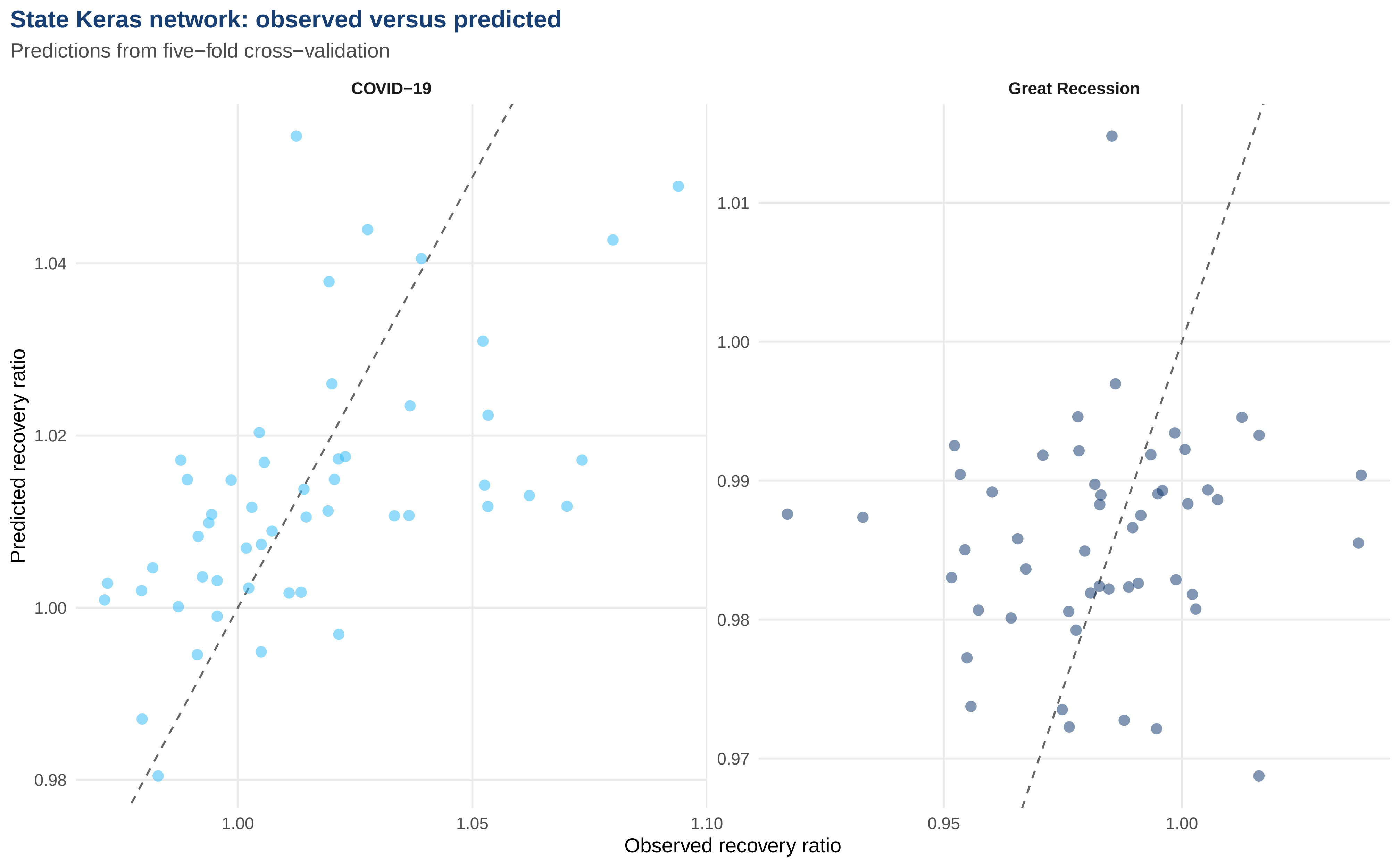
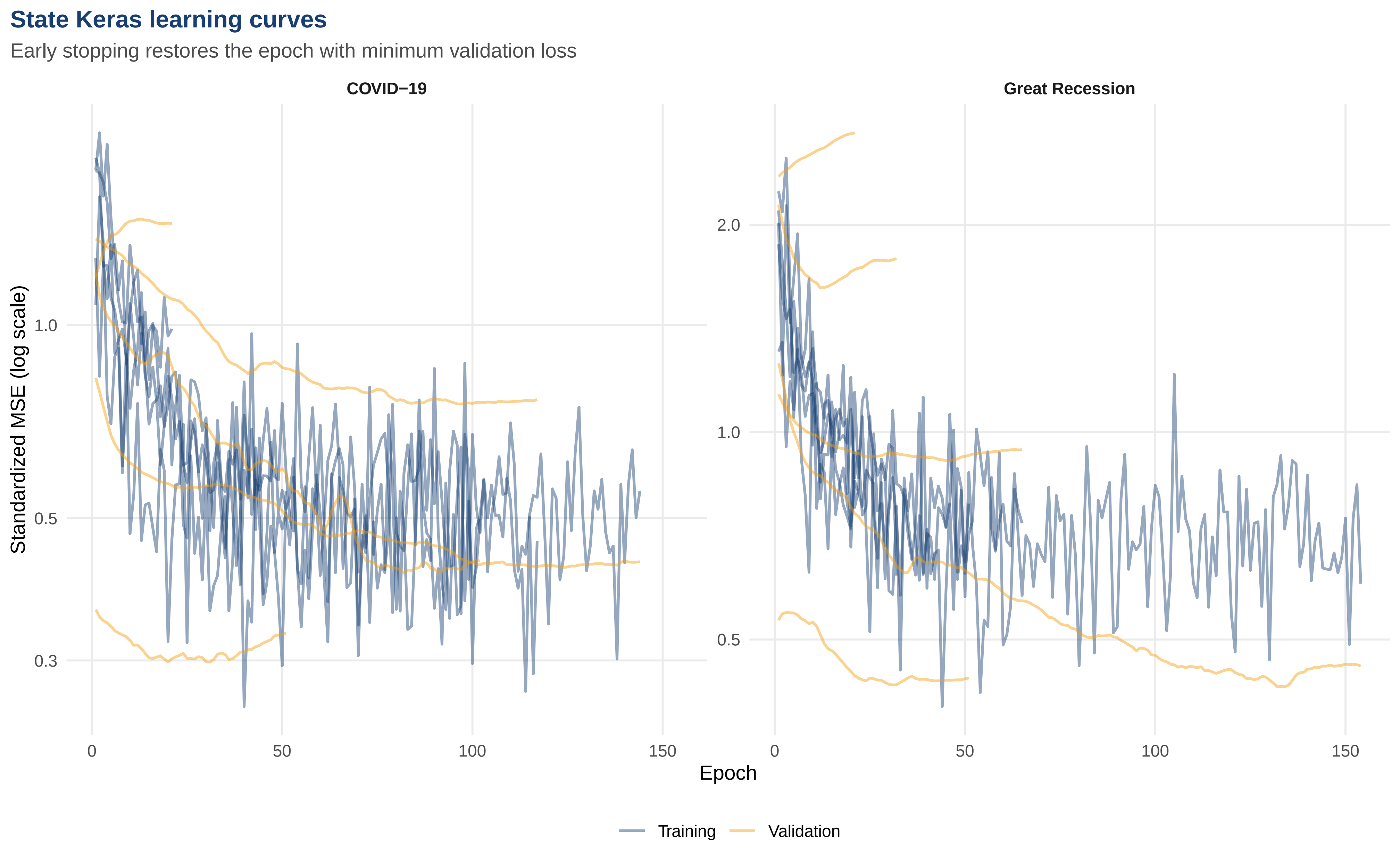
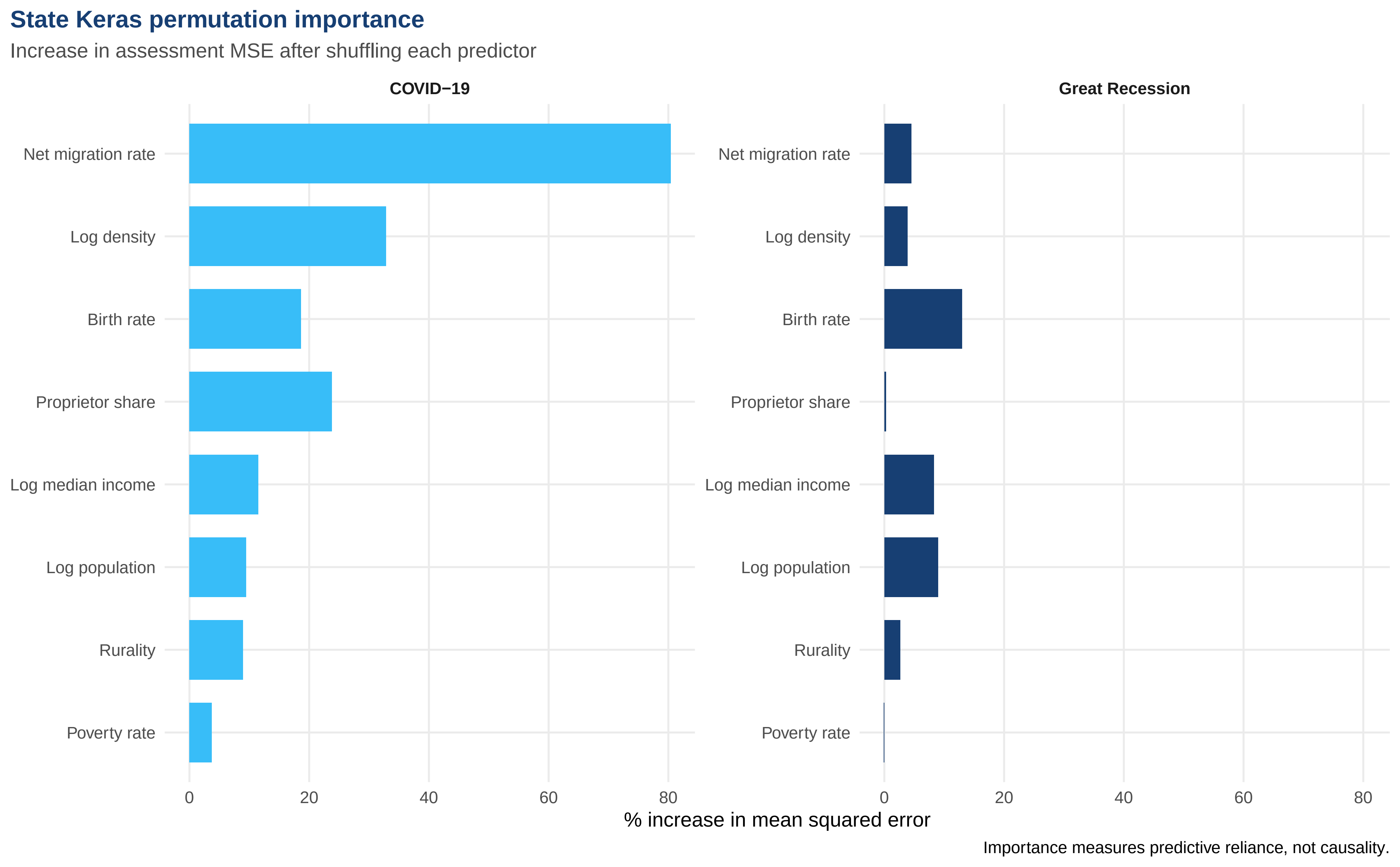
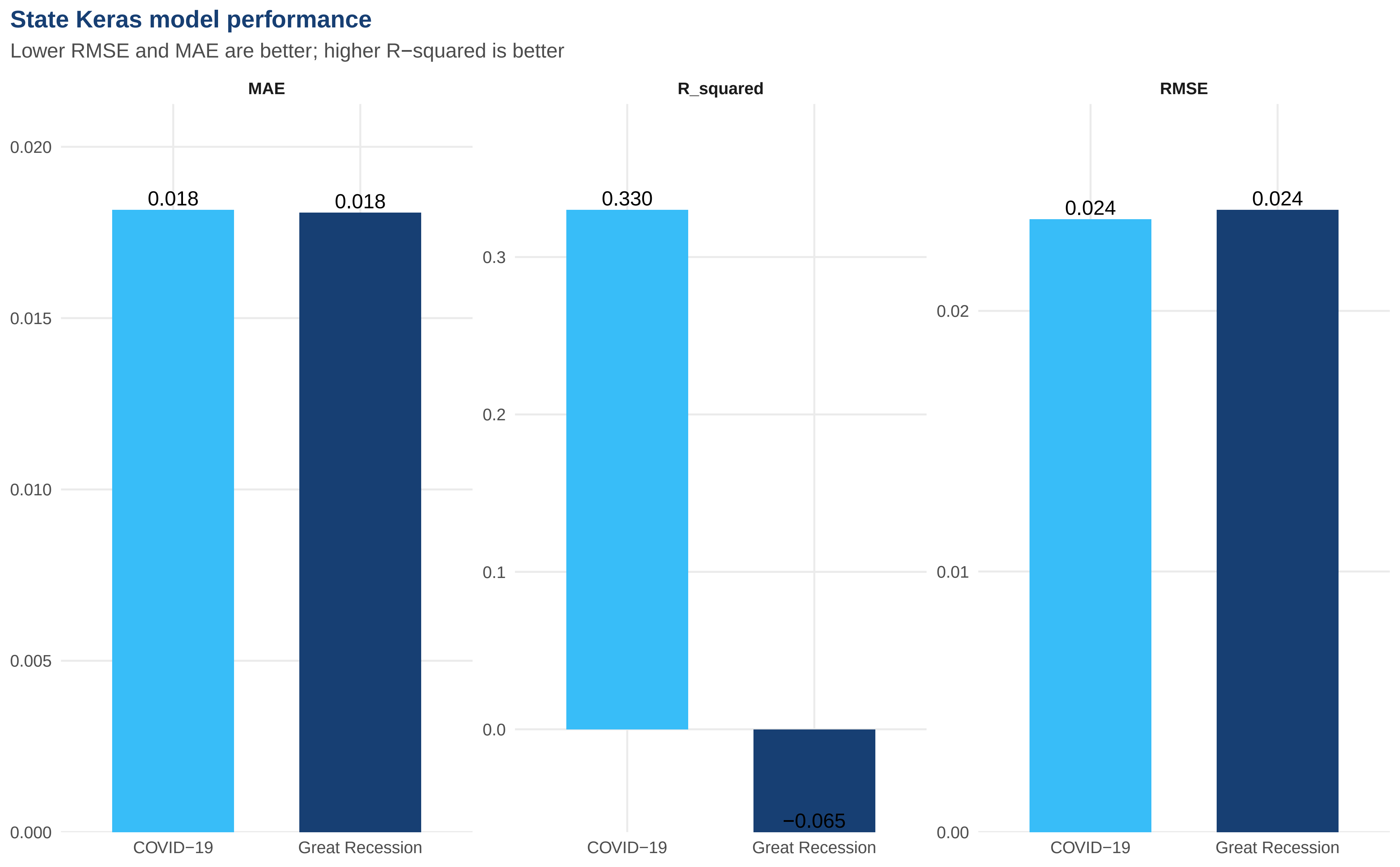

State Keras network: observed versus predicted
Predictions from five−fold cross−validation
COVID−19
Great Recession
Predicted recovery ratio
Observed recovery ratio
1.04
1.02
1.00
0.98
1.00
1.05
1.10
1.01
1.00
0.99
0.98
0.97
0.95
1.00


State Keras learning curves
Early stopping restores the epoch with minimum validation loss
COVID−19
Great Recession
Standardized MSE (log scale)
1.0
0.5
0.3
2.0
1.0
0.5
0
50
100
150
0
50
100
150
Epoch
Training
Validation



State Keras permutation importance
Increase in assessment MSE after shuffling each predictor
COVID−19
Great Recession
Net migration rate
Log density
Birth rate
Proprietor share
Log median income
Log population
Rurality
Poverty rate
0
20
40
60
80
% increase in mean squared error
Importance measures predictive reliance, not causality.


State Keras model performance
Lower RMSE and MAE are better; higher R−squared is better
MAE
R_squared
RMSE
0.018
0.018
0.330
−0.065
0.024
0.024
0.000
0.005
0.010
0.015
0.020
0.0
0.1
0.2
0.3
0.00
0.01
0.02
COVID−19
Great Recession
COVID−19
Great Recession
COVID−19
Great Recession

State Keras residual distributions

Residual = observed minus predicted recovery ratio